

GIT 16

-  **:tada:**
- Initial commit or project bootstrap.
-  **:bug:**
- Fix functional bugs in RTL, verification, Python, scripts, or flow code.
-  **:ambulance:**
- Fix urgent breakage: failing regression, broken CI, lab blocker, release blocker, or tapeout blocker.
-  **:sparkles:**
- Add a complete new feature.
-  **:construction:**
- Commit incomplete work intentionally.
-  **:boom:**
- Breaking interface, register map, protocol, API, timing contract, or CLI change.
-  **:recycle:**
- Refactor without intended behavior change.
-  **:art:**
- Improve structure, formatting, naming, or readability without behavior change.
-  **:fire:**
- Remove obsolete code, files, stale scripts, or generated artifacts.
-  **:coffin:**
- Remove dead signals, unreachable FSM states, unused functions, or obsolete helpers.
-  **:wastebasket:**
- Deprecate code or behavior that should be removed later.
-  **:truck:**
- Move or rename files, modules, packages, paths, or resources.
-  **:rewind:**
- Revert a previous change.
-  **:twisted_rightwards_arrows:**
- Merge branches. Do not use for muxing, routing, or arbitration.
-  **:see_no_evil:**
- Add or update `.gitignore`.
-  **:bookmark:**
- Release or version tag.

DOCS 4

-  **:memo:**
- Add or update docs, specs, register maps, integration notes, or README content.
-  **:bulb:**
- Add or update source comments explaining hardware behavior, assumptions, or constraints.
-  **:pencil2:**
- Fix typos in docs, comments, specs, or messages.
-  **:page_facing_up:**
- Add or update license.

TIMING 2

-  **:scissors:**
- Reduce combinational depth, retime logic, split paths, or improve Fmax.
-  **:stopwatch:**
- Benchmark instrumentation, cycle counting, latency measurement, or timing measurement infrastructure.

CI 2

-  **:construction_worker:**
- CI pipelines, GitHub Actions, build matrix, scheduled jobs, or regression automation.
-  **:green_heart:**
- Fix broken CI, regression automation, build matrix, or tool-cache issue.

RTL 12

-  **:bridge_at_night:**
- Clock-domain crossing, async boundary, synchronizer, handshake bridge, or reset-domain crossing.
-  **:rightwards_pushing_hand:**
- Valid/ready pushback, stall propagation, skid buffers, flow control, or downstream backpressure.
-  **:electric_plug:**
- Wire ports, buses, protocols, register interfaces, AXI/APB/TileLink/Wishbone connections, or top-level integration.
-  **:abacus:**
- Arithmetic, bit-width, fixed-point, saturation, rounding, accumulator, multiplier, divider, or datapath logic.
-  **:vertical_traffic_light:**
- Arbitration, priority, scheduling, traffic shaping, QoS, credits, or grant logic.
-  **:compass:**
- Address maps, memory maps, topology, routing tables, NoC coordinates, or integration direction.
-  **:dna:**
- Generics, parameters, SKUs, feature variants, configurable widths/depths, or generated RTL variants.
-  **:fire_extinguisher:**
- Fix X-prop, glitch risk, unsafe reset behavior, metastability risk, race, or silicon-escape-class hazard.
-  **:magnet:**
- Board-level work, IO, pins, PHYs, lab-only fixes, FPGA bring-up, or constraints tied to real hardware.
-  **:satellite_antenna:**
- PCIe, Ethernet, DDR, SerDes, SPI, I2C, UART, or other external protocol/link behavior.
-  **:safety_pin:**
- Small localized hardware workaround or mitigation for a known issue.
-  **:screwdriver:**
- Small mechanical RTL adjustment too narrow for a bigger category.

PYTHON 5

-  **:snake:**
- Python-specific implementation work when no more precise standard Gitmoji applies.
-  **:label:**
- Add or update Python type hints, dataclasses, TypedDicts, Pydantic models, SV typedefs, enums, or structs.
-  **:goal_net:**
- Improve exceptions, parser errors, failure handling, or user-facing error messages.
-  **:monocle_face:**
- Data exploration, report inspection, one-off analysis, or investigation scripts.
-  **:thread:**
- Threading, multiprocessing, async, job queues, or concurrent report processing.

INFRASTRUCTURE 2

-  **:bricks:**
- Hardware-flow infrastructure: runners, artifact storage, caches, job orchestration, or shared tool setup.
-  **:package:**
- Add or update generated bundles, packaged IP, release artifacts, reports, or bitstreams.

VERIFICATION 7

-  **:white_check_mark:**
- Add, update, or fix passing tests, regressions, assertions, formal checks, or checkers.
-  **:test_tube:**
- Add a failing test or reproducer for a known bug.
-  **:rotating_light:**
- Fix compiler, linter, CDC, synthesis, simulation, or timing warnings.
-  **:game_die:**
- Randomized testing, constrained-random stimulus, fuzzing, seeds, or randomized regressions.
-  **:microscope:**
- Add probes, traces, counters, waveform hooks, debug CSRs, or observability.
-  **:safety_vest:**
- Add validation, guards, illegal-state checks, protocol checks, or defensive checks.
-  **:seedling:**
- Add seed files, sample inputs, golden data, or deterministic regression data.

DEPENDENCIES 6

-  **:arrow_up:**
- Upgrade dependency, vendor IP, tool version, package, or submodule.
-  **:arrow_down:**
- Downgrade dependency, vendor IP, tool version, package, or submodule.
-  **:pushpin:**
- Pin exact versions of tools, IPs, packages, lockfiles, containers, or submodules.
-  **:heavy_plus_sign:**
- Add a dependency, package, included IP, submodule, or tool requirement.
-  **:heavy_minus_sign:**
- Remove a dependency, package, included IP, submodule, or tool requirement.
-  **:alien:**
- Update code due to external API, protocol, vendor IP, tool output, or report-format changes.

PPA 4

-  **:zap:**
- Improve speed, throughput, latency, benchmark result, or runtime.
-  **:chart_with_downwards_trend:**
- Reduce LUTs, FFs, gates, SRAM, BRAM, DSPs, cells, or general resource usage.
-  **:chart_with_upwards_trend:**
- Accept area, power, or resource increase intentionally for feature, timing, verification, or performance.
-  **:battery:**
- Clock gating, power gating, idle behavior, low-power mode, activity reduction, or power optimization.

SCRIPTS 2

-  **:hammer:**
- Build, sim, synth, lint, formal, regression, packaging, or helper scripts.
-  **:wrench:**
- Tool config, environment config, CLI config, YAML/TOML/JSON config, synthesis/sim flags, or parameters.